Phase-Adaptive Quantization

Design-Space Exploration of 4-Bit MMA Rescale Pipelines

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Prefill = compute-bound

Decode = memory-bound

Motivation

- **Existing 4-bit MMA formats** use one fixed rescale pipeline for both prefill and decode.
- **Prefill** is **compute-bound,** while decode is memory-bandwidth-bound.
- **For custom accelerators,** the rescale pipeline itself should be co-designed with the workload and the inference phase.

Underlying Principles

WA44 search space over 3 orthogonal axes

- Rescale levels *r*: 0 / 1 / 2
- *Tile block size b:* 16 / 32
- Precision choices: *scale format and accumulator precision*

10 representative configurations C0–C9

- NVFP4 = C1
- NVFP4-like = C7

Structured search space → Canonical rescale operators → AccelForge + Timeloop + Accelergy → Accuracy emulation

Search block granularily first, then pick the cheapest algebra within the admitted family.

Canonical Einsum View

Activation Tensor A (M×K) × *Weight Tensor W* (K×N) → Output Y (M×N)

BlockScale + Quantize | BlockScale + Quantize | Rescale Operators

$$Y = \text{einsum}('mk,kn->mn', A_q, W_q)$$

A_q = Quantize(BlockScale(A))

W_q = Quantize(BlockScale(W))

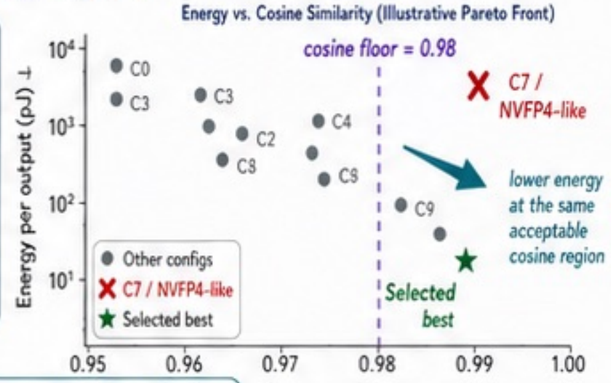
Evaluation

- **Accelerator:** 8 × 8 PE array, 64 KB GLB, 128 B RF/PE, 45 nm
- **Workloads:** LLM (Llama-2-FFN, VLM Qwen-2-VL ViT, VLA OpenVLA Head
- **Phases:** Prefill (M= 128), Decode (M=1)
- **Runs:** 60 hardware runs + 60 matched accuracy-emulation runs
- **Metric:** energy per output (pJ/output)
- *Accuracy floor:* cosine similarity ≥ 0.98

Results and Analysis of Results

A *Result 1: C7 / NVFP4-like is feasible, but never energy-optimal under cosine ≥ 0.98*

	Cell	Best	Saving vs C7
1	*LLM Prefill*	C1	58.1%
2	*LLM Decode*	C8	48.0%
3	*VLM Prefill*	C1	60.0%
4	*VLM Decode*	C1	50.4%
5	*VLA Prefill*	C9	50.7%
6	*VLA Decode*	C8	47.6%



C7 passes the accuracy floor, but in every workload-phase cell there exists another configuration with lower energy.

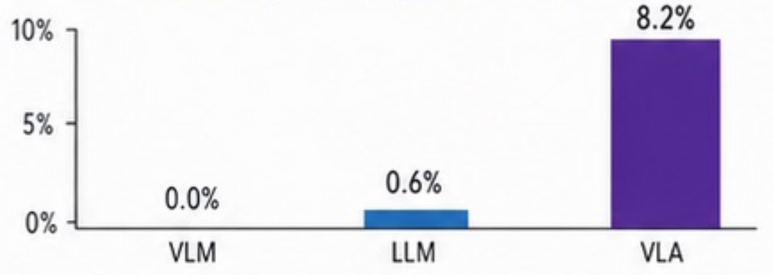
B *Result 2: Efficient two-step search*

1. **Step 1: Floor pass** — *Admit the coarsest block-granularity family that clears cosine ≥ 0.98*
2. **Step 2: Within-family cost pick** — *Choose the configuration with the lowest energy per output*

Matches exhaustive search on all 6 workload-phase cells.

C *Result 3: Phase-adaptive datapaths are conditional*

Benefit vs. best single fixed configuration at α = 0.5



A second silicon mode is most valuable for VLA, marginal for LLM, and unnecessary for VLM.

Best Configuration per Workload and Phase

	Cell	Phase	Best Config	Total Energy (pJ/output)	Cosine Similarity	Energy Savings vs NVFP4
1	LLM	Prefill	C1	5262	0.9843	58.1%
2	LLM	Decode	C8	7767	0.9807	48.0%
3	VLM	Prefill	C1	3952	0.9836	60.0%
4	VLM	Decode	C1	5557	0.9874	50.4%
5	VLA	Prefill	C9	6263	0.9895	50.7%
6	VLA	Decode	C8	7891	0.9950	47.6%

Why NVFP4 Loses

C1 total energy **7.41** GpJ vs. C7 total energy **17.71** GpJ

2.39× gap

The PF4 MAC datapath is identical; the gap is in rescale MAC and memory traffic.

LLM *Large Language Model (Llama-2-FFN)*
VLM *Vision-Language Model (Qwen-2-VL ViT)*
VLA *Vision-Language-Action (OpenVLA Head)*

- *Prefill batch dimension (prompt length) M = 128*
- *Decode batch dimension M = 1*
- *Results averaged over 60 hardware runs and 60 matched accuracy-emulation runs.*

References

[1] *NVIDIA, "Introducing NVFP4: Efficient and Accurate Low-Precision Inference," 2025.*

[2] *Open Compute Project, "Microscaling Formats (MX) Specification 1.0," 2023.*

[3] *Rouhani et al., "Microscaling Data Formats for Deep Learning," 2023.*

[4] *Fantar et al., "GPTQ," ICLR 2023.*

[5] *Lin et al., "AWQ," MLSys 2024.*

[6] *Xiao et al., "SmoothQuant," ICML 2023.*

[7] *Parashar et al., "Timeloop," ISPASS 2019.*

[8] *Wu et al., "Accelergy," ICCAD 2019.*

[9] *Zhong et al., "DgSeme," OSDI 2024.*

[10] *Patel et al., "Spilliinis," ISCA 2024.*

Repository: github.com/ETB23282/Hardware-for-quantization